

[36] Human Placenta Thioredoxin Reductase: Preparation and Inhibitor Studies

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Introduction

Mammalian thioredoxin reductases [EC 1.6.4.5; $\text{NADPH} + \text{H}^+ + \text{thioredoxin-S}_2 \rightleftharpoons \text{NADP}^+ + \text{thioredoxin-(SH)}_2$] is a dimeric (M_r 55,000 per subunit) flavo-enzyme. As a pyridine nucleotide-disulfide oxidoreductase it is closely related to glutathione reductase.¹⁻³ All members of this enzyme family have FAD and an adjacent redox active disulfide, but mammalian thioredoxin reductases contain, as a common feature, an additional redox-active center in the form of a penultimate Cys-Sec sequence.^{4,5} This additional redox center is located on a flexible C-terminal extension, which explains another characteristic of large thioredoxin reductases—their unusual broad substrate specificity.⁶ This aspect might explain a large variety of intracellular processes, such as electron delivery to ribonucleotide reductase but also extracellular processes, such as cytokine effects, that are based on the activities of thioredoxin reductase (TrxR) and its major substrate thioredoxin (Trx).⁷⁻⁹

Many studies on TrxR have been carried out with small TrxRs of bacterial origin. These enzymes, however, differ fundamentally from the large TrxRs^{2,3} in size, structure, substrate spectrum, and catalytic mechanism. Furthermore, because of the broad substrate spectrum of large TrxRs, many *in vitro* findings may lead to functional misinterpretations, particularly because *in vivo* data are still rare. It is our goal to identify probes that selectively modify thioredoxin reductase and that can be employed as enzyme inhibitors both *in vitro* and *in vivo*. From the enzymologist's point of view many different inhibitors with clear cut effects on the intermediate

¹ A. Holmgren, *J. Biol. Chem.* **252**, 4600 (1977).

² L. D. Arscott, S. Gromer, R. H. Schirmer, K. Becker, and C. H. Williams, Jr., *Proc. Natl. Acad. Sci. U.S.A.* **94**, 3621 (1997).

³ C. H. Williams, L. D. Arscott, S. Müller, B. W. Lennon, M. L. Ludwig, P. F. Wang, D. M. Veine, K. Becker, and R. H. Schirmer, *Eur. J. Biochem.* **267**, 6110 (2000).

⁴ V. N. Gladyshev, K. T. Jeang, and T. C. Stadtman, *Proc. Natl. Acad. Sci. U.S.A.* **93**, 6146 (1996).

⁵ S. Gromer, R. H. Schirmer, and K. Becker, *FEBS Lett.* **412**, 318 (1997).

⁶ S. Gromer, J. Wissing, D. Behne, K. Ashman, R. H. Schirmer, L. Flohé, and K. Becker, *Biochem. J.* **332**, 591 (1998).

⁷ S. Gromer, R. H. Schirmer, and K. Becker, *Redox Rep.* **4**, 221 (1999).

⁸ D. Mustacich and G. Powis, *Biochem. J.* **346**, 1 (2000).

⁹ E. S. J. Arnér and A. Holmgren, *Eur. J. Biochem.* **267**, 6102 (2000).

states of the catalytic cycle are desirable. Physiological studies would benefit from highly selective, well-controllable inhibitors. Bearing in mind the medical implications of the enzyme (e.g., in tumor therapy¹⁰) such inhibitors should provide us with pharmacological lead substances—a prerequisite for rational drug design.

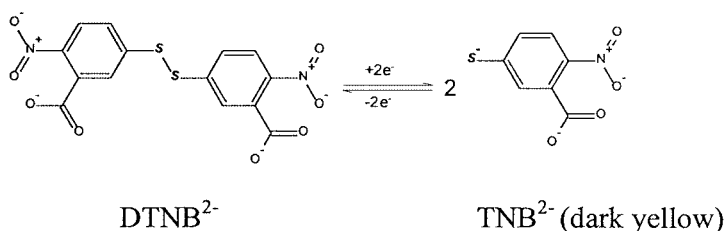
To assess the effects of potential inhibitors on enzyme activity a highly purified enzyme must be available not only for kinetic and X-ray crystallography studies but also as a reference for experiments on cell cultures. Enzyme preparations from native tissue are still necessary as the novel recombinant techniques for selenoenzymes^{11,12} such as TrxR are promising but do not yet produce satisfying material.

Assay Procedures

Several different assays for the determination of thioredoxin reductase activity have been developed.¹³ They differ in specificity, cost, and complexity. Here we present the most frequently applied spectrophotometric assays and comment on their specific properties. All assays are carried out at 25°.

5,5'-Dithiobis(2-Nitrobenzoic Acid) Reduction Assay

The 5,5'-dithiobis(2-nitrobenzoic acid) (DTNB) reduction assay¹³ is based on the broad substrate specificity of large thioredoxin reductases, which are capable of reducing DTNB (Ellman's reagent) to form two molecules of 2-nitro-5-thiobenzoate anion (TNB).



Because the thiolate of TNB is mainly responsible for the high absorbance at 412 nm the pH is of importance and should be above pH 7.0. Most publications utilize an $\varepsilon_{\text{TNB}, 412 \text{ nm}}$ of $13.6 \text{ mM}^{-1} \text{ cm}^{-1}$ for TNB. This value is not quite correct

¹⁰ K. Becker, S. Gromer, R. H. Schirmer, and S. Müller, *Eur. J. Biochem.* **267**, 6118 (2000).

¹¹ E. S. J. Arnér, H. Sarioglu, F. Lottspeich, A. Holmgren, and A. Böck, *J. Mol. Biol.* **292**, 1003 (1999).

¹² S. Bar-Noy, S. N. Gorlatov, and T. C. Stadtman, *Free Radic. Biol. Med.* **30**, 51 (2001).

¹³ E. S. J. Arnér, L. Zhong, and A. Holmgren, *Methods Enzymol.* **300**, 226 (1999).

(approximately ± 1) under the conditions applied¹⁴ but should be used in order to keep results comparable.

Unlike earlier protocols, we omit bovine serum albumin (BSA) in the buffer because BSA reacts with DTNB and might interfere with enzyme activity* and inhibitors. Although dilute human TrxR is much more stable in the presence of NADPH than other members of the disulfide reductase family, it should be noted that the loss of enzyme activity is approximately 50% after 1 hr when (pure) human TrxR is incubated at 1 nM concentration in assay buffer with 200 μM NADPH at 25°. Inhibitors that are not easily water soluble should be dissolved in dimethyl sulfoxide (DMSO) if possible. DMSO up to 10% (v/v) in the assay mixture does not alter the enzyme activity and is easier and more reliable to pipette than, for instance, ethanol.

Reagents

T-buffer: 100 mM Potassium phosphate, 2 mM EDTA; pH 7.4

DTNB: 100 mM DTNB in DMSO (store frozen, keep dark)

NADPH: 4 mM NADPH in T-buffer (prepare freshly every day, keep on ice)

In a total assay volume of 1 ml, 30 μl of DTNB (to 3 mM) is mixed with T-buffer, enzyme sample ($\sim 1\text{--}2$ nM), and if applicable inhibitor solution. The enzymatic reaction is initiated by adding 50 μl of NADPH (to 200 μM). As the spectrophotometric signal of the reaction is usually not linear, particularly not at low enzyme concentrations, the TNB-dependent increase in absorbance at 412 nm should be monitored during the “most linear” phase within the first 3 min. For inhibition studies it is also helpful to state the time interval for which the $\Delta A \text{ min}^{-1}$ is calculated (e.g., the range between 20 and 80 sec) in order to increase the reproducibility of the results. As 1 unit of enzymatic activity is defined as the consumption of 1 μmol of NADPH per minute (meaning the production of two TNB per NADPH), activity in the assay mixture ($V = 1$ ml; $d = 1$ cm) is calculated as follows:

$$\text{Volume activity} = \frac{\Delta A_{412 \text{ nm}}}{1 \text{ min} \cdot 2 \cdot 0.0136 \mu\text{M}^{-1} \cdot 1} \left[\frac{\mu\text{M}}{\text{min}} \right] \quad \text{or} \quad \left[\frac{\text{mU}}{\text{ml}} \right] \quad (1)$$

The DTNB reduction assay is inexpensive, simple, and (because of the high ε_{412} value of TNB and the stoichiometry of the reaction) sensitive; it lacks, however, specificity. For instance, glutathione reductase can also reduce DTNB to give

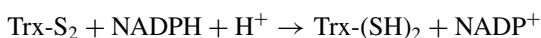
* Increase up to 25%.

¹⁴ P. W. Riddles, R. L. Blakeley, and B. Zerner, *Methods Enzymol.* **91**, 49 (1983).

TNB¹⁵ (although with a low K_{cat} of $\sim 500 \text{ min}^{-1}$). Also, free SH groups (which are often present in crude extracts in relevant concentrations) react nonspecifically with DTNB to form mixed disulfides and TNB.¹⁴ Furthermore, studies with non-selenium-containing large TrxRs¹⁶ and selective proteolytic digests^{6,17} of human TrxR indicate that DTNB may, unlike thioredoxin, be reduced not only at the C-terminal Cys–Sec redox center but also at the redox-active disulfide of the protein (Cys-57 and Cys-62).

Thioredoxin Reduction Assay

This is the most specific but also the most expensive assay system.



Native human thioredoxin (hTrx) causes technical difficulties due to dimer formation and because hTrx mutants lacking dimer formation are not yet affordable in the required amounts (IMCO, Stockholm, Sweden), *Escherichia coli* thioredoxin is often used for the reaction.¹⁸ Thioredoxins of other species such as *Plasmodium falciparum* Trx exhibit low K_m values and protocols for rapid preparation are available. They should be considered as alternatives.^{19,20}

Reagents

T-buffer: 100 mM Potassium phosphate, 2 mM EDTA; pH 7.4

Trx: 5 mM oxidized *E. coli* thioredoxin [$\epsilon_{280 \text{ nm}}$ (*E. coli* Trx-S₂) = $13.7 \text{ mM}^{-1} \text{ cm}^{-1}$] in T-buffer (store frozen, keep on ice)

NADPH: 4 mM NADPH in T-buffer (prepare freshly every day, keep on ice)

In a total assay volume of 1 ml, 25 μl of NADPH (to 100 μM) is mixed with T-buffer and enzyme sample (approximately 3 nM). The enzymatic reaction is initiated by adding 25 μl of (*E. coli*) Trx stock solution (to 125 μM) and the decrease in absorbance at 340 nm is monitored. One unit of enzyme activity is defined as the consumption of 1 μmol of NADPH per minute ($V = 1 \text{ ml}$; $d = 1 \text{ cm}$).

$$\text{Volume activity} = \frac{\Delta A_{340 \text{ nm}}}{1 \text{ min} \cdot 0.00622 \mu\text{M}^{-1} \cdot 1} \left[\frac{\mu\text{M}}{\text{min}} \right] \quad \text{or} \quad \left[\frac{\text{mU}}{\text{ml}} \right] \quad (2)$$

¹⁵ K. K. Wong, M. A. Vanoni, and J. S. Blanchard, *Biochemistry* **27**, 7091 (1988).

¹⁶ T. W. Gilberger, B. Bergmann, R. D. Walter, and S. Müller, *FEBS Lett.* **425**, 407 (1998).

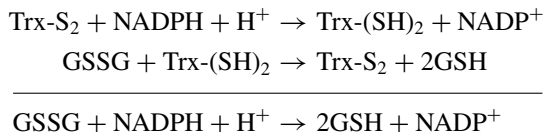
¹⁷ L. Zhong, E. S. J. Arnér, J. Ljung, F. Åslund, and A. Holmgren, *J. Biol. Chem.* **273**, 8581 (1998).

¹⁸ S. B. Mulrooney, *Protein Expr. Purif.* **9**, 372 (1997).

As an alternative to *E. coli* Trx (K_m with human TrxR = 25 μM), recombinant thioredoxin of the malarial parasite *P. falciparum*¹⁹ (K_m with hTrxR = 2 μM) can be used for assaying hTrxR activity. *Pf*Trx does not show dimer formation disturbing the reaction, it can be easily overexpressed in *E. coli*, purified by affinity chromatography, and—because of its low K_m with human TrxR—20 μM *Pf*Trx produces a stable spectrophotometric signal under substrate saturation over minutes.

GHOST Assay*

Despite its broad substrate specificity thioredoxin reductase is incapable of reducing glutathione disulfide (GSSG). Its product thioredoxin, in the reduced state, however, easily reduces GSSG to form reduced glutathione (GSH). Apart from its potential physiological importance, this novel assay system is useful to measure thioredoxin reductase activity in samples in which, for example, other assay constituents such as inhibitors interfere with the TNB determination at 412 nm. Furthermore, this assay yields a better linear signal than the DTNB reduction assay and it requires only catalytic concentrations of Trx.¹⁹



As the overall reaction is also catalyzed by glutathione reductase, the presence of this enzyme can complicate the assay. Thus the functional absence of this enzyme is a prerequisite of this assay.

Reagents

T-buffer: 100 mM Potassium phosphate, 2 mM EDTA; pH 7.4

Trx: 1 mM *E. coli* thioredoxin in T-buffer (store frozen, keep on ice)

NADPH: 4 mM NADPH in T-buffer (prepare freshly every day, keep on ice)

GSSG: 20 mM Glutathione disulfide in T-buffer (store frozen, keep on ice)

Procedure. In a total assay volume of 1 ml, 25 μl of NADPH (to 100 μM) is mixed with T-buffer, 30 μl of Trx stock solution (to 30 μM), and enzyme sample (approximately 1–3 nM). The enzymatic reaction is initiated by adding 50 μl of GSSG (to 1 mM). If the assay mixture should still contain a small amount

* GHOST, Glutathione as substrate of thioredoxin.

¹⁹ S. M. Kanzok, R. H. Schirmer, I. Turbachova, R. Iozef, and K. Becker, *J. Biol. Chem.* **275**, 40180 (2000).

²⁰ S. M. Kanzok, A. Fechner, H. Bauer, J. K. Ulschmid, H. M. Müller, J. Botella-Munoz, S. Schneuwly, R. H. Schirmer, and K. Becker, *Science* **291**, 643 (2001).

of glutathione reductase its share of the overall activity can be calculated and subtracted if GSSG is added before Trx; in other words, if Trx is used to start the assay.

One unit of enzymatic activity is defined as the consumption of 1 μ mol of NADPH per minute. For activity calculations, see Eq. (2).

Instead of *E. coli* Trx, also thioredoxins of other species can be employed in this assay.¹⁹

Note that this assay system is influenced by the concentrations of GSSG, Trx, and TrxR. It should also be pointed out that thioredoxin stock solutions are often stored in the presence of reductants such as dithioerythritol (DTE). This “contamination” may lead to glutathione and thioredoxin disulfide consumption to relevant amounts. This would result in inaccurate enzyme activity measurements.

Auranofin Inhibition Assay of Biological Samples

Hill *et al.*²¹ published a simple and effective method to determine thioredoxin reductase activity in biological samples by exploiting the inhibitory effects of gold thioglucose (aurothioglucose). In two parallel DTNB reduction assays DTNB reduction is determined, the first being a standard assay. The second, “non-TrxR” assay contains 10 μ M aurothioglucose, leading to TrxR inhibition. The difference in activity between the two assays represents TrxR activity. We have modified this procedure by using auranofin instead of gold thioglucose. Auranofin is more stable in solution (dimethyl sulfoxide, DMSO) than aurothioglucose and lower concentrations are required for a specific and complete inhibition²² of human TrxR. The mechanistically related enzyme glutathione reductase and the selenium-containing glutathione peroxidase are by orders of magnitude less susceptible to this inhibition. Because the Au(I)-induced TrxR inhibition is virtually not reversible by dialysis, an NADPH preincubation with and without auranofin for 5 min and consecutive dialysis can be performed. This could reduce the risk of inhibition of enzymes other than TrxR that contribute to the overall DTNB reduction. Otherwise an overestimation of TrxR activity is theoretically possible. However, the limitations of the DTNB reduction assay should be accounted for when interpreting the results.

Reagents

T-buffer: 100 mM Potassium phosphate, 2 mM EDTA; pH 7.4

DTNB: 100 mM DTNB in DMSO (store frozen, keep dark)

Auranofin: 1 mM in DMSO (store cold, keep dark)

NADPH: 4 mM NADPH in T-buffer (prepare freshly every day, keep on ice)

²¹ K. E. Hill, G. W. McCollum, and R. F. Burk, *Anal. Biochem.* **253**, 123 (1997).

²² S. Gromer, L. D. Arscott, C. H. Williams, Jr., R. H. Schirmer, and K. Becker, *J. Biol. Chem.* **273**, 20096 (1998).

Assay 1

In a total assay volume of 1 ml, 30 μ l of DTNB is mixed with buffer and enzyme sample (approximately 1–2 nM). The enzymatic reaction is initiated by adding 50 μ l of NADPH (to 200 μ M).

Assay 2: Non-TrxR

The procedure is the same as in assay 1, but 1 μ l of buffer is replaced by 1 μ l of auranofin solution (to 1 μ M).

By using $\Delta A \text{ min}^{-1} = \Delta A \text{ min}^{-1} \text{ of assay 1} - \Delta A \text{ min}^{-1} \text{ of assay 2}$ in Eq. (1), TrxR activity can be calculated. This assay can also be carried out with Trx-S₂ as oxidizing substrate instead of DTNB.

Enzyme Purification

The TE-buffer used throughout the preparation consists of 50 mM Tris-HCl, 1 mM EDTA, pH 7.6. In the method described here,²² human placenta was used as starting material. As judged by SDS-PAGE, cello gel electrophoresis, kinetic studies and spectroscopic experiments this procedure yields an extremely pure and homogeneous enzyme. The procedure can be applied to other tissues and species, however, the elution conditions may however vary.

Chloroform–1-Butanol Extraction. Placenta tissue is weighed out and homogenized in a Waring blender with 0.6 ml of extraction solution [10 μ M FAD and 40 μ M phenylmethylsulfonyl fluoride (PMSF) in TE buffer] per gram of placenta. If the tissue is frozen it should be thawed in plastic bags together with the extraction buffer by placing the closed bags in a 40° water bath before homogenization. Immediately before treatment with chloroform–1-butanol²³ (which should be carried out in a hood), the homogenate is titrated to pH 8.3 with 5 M NH₄OH. The –20° cold chloroform–1-butanol mixture (1 : 2.5, v/v; 120 μ l/g of placenta) is added under vigorous stirring. It is important to rehomogenize the brownish suspension in the Waring blender. The resulting suspension is left for 30 min to 1 hr on ice and then centrifuged at $\geq 8000g$ for 90 min at 4°. The supernatant is filtered through glass wool and adjusted to pH 8.3 with 5 M NH₄OH. If the used tissue is limited the precipitate can be taken up in extraction solution (0.4 ml/g of placenta), homogenized, and centrifuged as described above and the supernatants are combined accordingly.

Acetone Precipitation. Per 1 ml of chloroform–butanol extract, 0.85 ml of ice-cold acetone is slowly added under stirring. The solution is left for approximately 1 hr on ice and then centrifuged at $\geq 3500g$ for 15 min at 4°. The clear, yellow supernatant is discarded. In bulk preparations when high-rpm centrifuges with large capacity become a limiting factor, the supernatant sometimes appears cloudy but hardly ever contains relevant amounts of activity. When in doubt a quick DTNB

²³ E. M. Scott, *Prep. Biochem.* **6**, 147 (1976).

reduction assay should be performed. The gluey brownish pellet is carefully taken up with a glass bar by adding small volumes of TE buffer to a final volume of approximately 100 ml. This sample is dialyzed exhaustively against 2-fold-diluted TE buffer (25 mM Tris-HCl, 0.5 mM EDTA, pH 7.6) overnight and centrifuged (30 min, $\geq 25,000g$, 4°). The supernatant is set aside while the pellet is resuspended in TE buffer, mixed carefully, and centrifuged as described above. The combined supernatants are filtered through glass wool if gross impurities should persist and adjusted to pH 8.3 with 5 M NH_4OH .

DEAE-52 Cellulose Chromatography. The acetone-treated fraction is slowly applied at room temperature to a DEAE-52 cellulose column (Whatman, Clifton, NJ) equilibrated with TE buffer (approximately 0.6-ml final column volume per gram of placenta is required). After washing the column with 2 column volumes of TE buffer followed by 1 volume of 50 mM NaCl in TE buffer, thioredoxin reductase activity is eluted with 90 mM NaCl in TE buffer. In this step the enzyme comigrates with a deep red protein. The pool of active fractions is concentrated and washed with TE buffer in a Centriprep 30 (Amicon, Danvers, MA). The resulting solution is diluted 2-fold with TE buffer and the pH is adjusted to pH 7.6 with 100 mM HCl.

2',5'-ADP-Sepharose 4B Affinity Chromatography. The above-described fraction is applied to a 30-ml (1.5×17 cm) 2',5'-ADP-Sepharose 4B column (Pharmacia, Piscataway, NJ) in a jacketed chromatography tube. The tube is cooled to $6 \pm 1^\circ$, the exact temperature being crucial for the purification success. The column is consecutively washed with 60 ml of TE buffer, 30 ml of 100 mM KCl in TE buffer, 20 ml of 200 mM KCl in TE, 30 ml of 100 mM KCl in TE, 60 ml of 2-fold-diluted TE, 60 ml of 500 μM NADH in TE, 60 ml of TE, 60 ml of 100 μM NADP^+ in TE, and 30 ml of 300 μM NADP^+ in TE. Finally, TrxR activity is eluted with 750 μM NADP^+ in TE buffer, concentrated, and washed with the buffer in a Centriprep 30. This solution, contains—on the basis of absorption spectra, specific activity, and sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS–PAGE) analysis—more than 95% pure thioredoxin reductase. Because the remaining impurities contain small amounts of glutathione reductase a further purification step may be warranted.

Sephadex G-200 Gel Filtration. The above described fraction is applied to a Sephadex G-200 column equilibrated at 4° with TE buffer (Pharmacia; 1×100 cm) in a jacketed chromatography tube. Active and glutathione reductase-free fractions are pooled and concentrated. The enzyme solution is stored at 4° . $(\text{NH}_4)_2\text{SO}_4$ precipitation results in a tremendous loss of activity and is therefore not recommended.

Inhibitor Studies

A selection of TrxR inhibitors is shown in Table I.^{24–32} Different methods can be applied to study a potential inhibitor. The easiest way is to add the inhibitor to

²⁴ K. U. Schallreuter, F. K. Gleason, and J. M. Wood, *Biochim. Biophys. Acta* **1054**, 14 (1990).

TABLE I
INHIBITORS OF MAMMALIAN THIOREDOXIN REDUCTASES^a

Inhibitor	Inhibition			Comment	Refs.
	NADPH dependent	Reversible	Irreversible		
Carmustine (BCNU) and other nitrosoureas	+	—	+	Effective inhibitors; however, not selective because mechanistically related enzymes such as glutathione reductase are also inactivated and DNA is alkylated	2, 5, 24
Azelaic acid (dicarboxylic acids)	NA	NA	NA	Original data were obtained with <i>E. coli</i> TrxR. No inhibitory effects on hTrxR by up to 1 mM azelaic acid found	25
Adriamycin and other anthracyclines	(+)	NA	NA	Original work was done with rat enzyme. Results were not reproducible	26, 27
1-Chloro-2,4-dinitrobenzene (CDNB)	+	—	+	2,4-DNP-arylation of C-terminal Cys and Sec. Induces considerable oxidase activity	28–30, this report
Auranofin and aurothioglucose	+	—	+	Selective tight-binding inhibitors. Effect can be reversed by the reducing chelator BAL	21, 22
Alkyl-2-imidazolyl disulfide analogs	—	+	—	Mainly competitive inhibitors with K_i values in lower micromolar range. Some are metabolized by enzyme	31
(2,2' : 6',2''-Terpyridine)platinum(II) complexes	+	+	+	New platinum(II) derivatives with reversible and irreversible inhibitory component. IC ₅₀ values as low as 2 nM, that is in stoichiometric concentrations	32

^a +, Yes; —, no; NA, not applicable.

the assay system and compare the activity with an untreated control. To further elucidate the mode of action but also to evaluate apparently weak inhibitors or covalent inhibitors that modify a protein time dependently, a preincubation protocol may be required. For members of the disulfide reductase family, the enzyme must often be in a reduced state before effective inhibition (e.g., by modification of the active site thiols) can take place. Because NADPH is an ingredient of all kinetic assay systems used, it is difficult to determine whether a specific substance does inhibit the oxidized, the reduced, or all forms of the enzyme. This problem can be overcome by incubating the enzyme in the absence and presence of NADPH with or without inhibitor; as stated above, native human TrxR is stable when exposed to NADPH. A sample of the preincubation solution is taken and assayed as usual (precaution must be taken to rule out that the sample carries over too much inhibitor into the assay vial, leading to additional inhibition). This procedure also allows the investigator to address the time and concentration dependence of the inhibition.

Example: Preincubation with Auranofin

In a final volume of 50 μl of T-buffer, 200 μM NADPH, 0.5 μM subunits TrxR, and 1 μM auranofin [1 μl of a 50- μl stock solution in dimethyl sulfoxide (DMSO)] are mixed at 25°. A corresponding control in which auranofin has been replaced by an equivalent volume of DMSO is also prepared.

Samples (3 μl) of each preincubation mixture are taken and activity is determined in separate DTNB reduction assays. This can be done at different time points if the inhibitor acts slower than auranofin. It is necessary to rule out that the transferred inhibitor concentration (here approximately 3 nM in the assay) has inhibitory effects.

Spectroscopy of Enzyme in Presence of Inhibitor. Mammalian TrxRs are flavo-enzymes. Thus, during the catalytic cycle typical spectroscopic changes can be monitored.² Inhibitors can bind at different sites of the enzyme and may lead to a block in these spectroscopic events or even result in a modification of the signal. Because the spectroscopic changes of human TrxR closely resemble those of glutathione reductase it is justified to make use of the information gathered for the latter enzyme.²

²⁵ K. U. Schallreuter and J. M. Wood, *Cancer Lett.* **36**, 297 (1987).

²⁶ B. L. Mau and G. Powis, *Free Radic. Res. Commun.* **8**, 365 (1990).

²⁷ B. L. Mau and G. Powis, *Biochem. Pharmacol.* **43**, 1613 (1992).

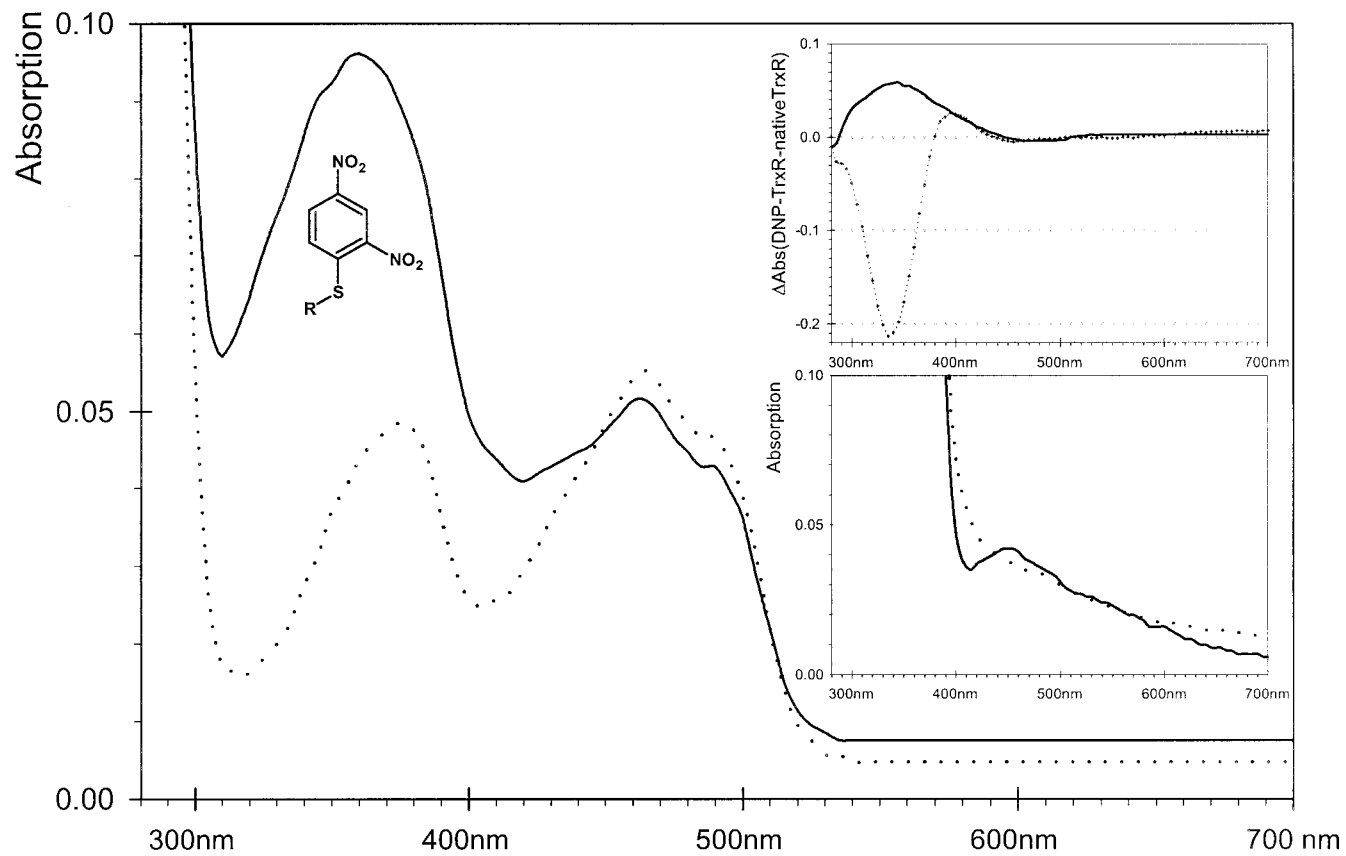
²⁸ E. S. J. Arnér, *Biofactors* **10**, 219 (1999).

²⁹ J. Nordberg, L. Zhong, A. Holmgren, and E. S. Arnér, *J. Biol. Chem.* **273**, 10835 (1998).

³⁰ E. S. J. Arnér, M. Björnstedt, and A. Holmgren, *J. Biol. Chem.* **270**, 3479 (1995).

³¹ J. E. Oblong, E. L. Chantler, A. Gallegos, D. L. Kirkpatrick, T. Chen, N. Marshall, and G. Powis, *Cancer Chemother. Pharmacol.* **34**, 434 (1994).

³² K. Becker, C. Herold-Mende, J. J. Park, G. Lowe, and R. H. Schirmer, *J. Med. Chem.* **44**, 2784 (2001).



Modifying groups with their own typical absorption patterns may be studied per se. An example is shown in Fig. 1.^{28,29,33} Changes in absorbance due to the microenvironment must be taken into consideration.

The inhibition of an enzyme can be reversible or irreversible. Thus the investigator should check the possible reversal of inhibition by dialysis or dilution, incubation with reducing agents [dithioerythritol (DTE), Cleland's reagent] or [in the case of metal–organic inhibitors such as gold(I) and platinum(II) complexes] by chelating agents²² such as British-Anti-Lewisite (BAL; 2,3-dithiopropional).

It is important to note, that reducing agents (DTE and BAL) interfere with the assay systems. The DTNB reduction assay, for instance, can be used only if these reducing agents are removed by gel filtration, dialysis, or chemical reactions in order to reduce the formation of nonenzymatically formed TNB. If the GHOST assay is employed the amount of transferred reducing equivalents can be calculated, and nonspecifically consumed GSSG can be replaced if necessary. GSH and GSSG (unlike higher concentrations of TNB) do not significantly interfere with the spectroscopic activity determination.

Apart from the spectroscopic methods to determine the site of action of a particular inhibitor, studies with mutant enzymes lacking the C-terminal Sec (e.g., Sec-496 → Cys) may provide additional information.^{16,34,35}

Crystallization

The crystallization of mammalian thioredoxin reductases remains a challenge. We have screened a large variety of crystallization conditions, using the hanging drop method.³⁶ The most promising results have been obtained at 25° with 5 μ l of protein solution human TrxR at (10 mg ml⁻¹ in 2 mM EDTA, pH 7.4) mixed with 5 μ l of 7% (w/v) polyethylene glycol (PEG) 8000 in 100 mM Tris-HCl, pH 7.4, in the drop. The reservoir contained 800 μ l of 12% (w/v) PEG 8000 in 100 mM

FIG. 1. Spectral changes of human TrxR due to a covalently bound inhibitor. In this example, human TrxR in the presence of NADPH is modified by 1-chloro-2,4-dinitrobenzene (CDNB). Two enzyme samples were incubated with CDNB in the presence and absence of 200 μ M NADPH for 1 hr. The samples were dialyzed thereafter and the spectra recorded. In the large diagram the absorption spectra of CDNB-modified (solid curve) and unmodified (dotted curve) human placenta TrxR are shown. The strong increase in absorbance between 300 and 400 nm is typical of an S-2,4-dinitrophenyl group [M. Bilzer, R. L. Krauth-Siegel, R. H. Schirmer, T. P. Akerboom, H. Sies, and G. E. Schulz, *Eur. J. Biochem.* **138**, 373 (1984)] (see *Inset* formula). *Lower inset*: Spectra for both enzyme forms in the presence of 200 μ M NADPH. It indicates that the charge transfer complex between the proximal thiolate (Cys-62) and FAD is still generated in the modified enzyme. *Upper inset*: Differential spectra of native and CDNB-modified TrxR as calculated from the other spectra are shown. The solid curve represents the oxidized form of the enzyme (modified TrxR – native TrxR). Assuming an $\epsilon_{340\text{ nm, Cys-2,4-DNP}}$ of 9.6 mM⁻¹ cm⁻¹, a binding of approximately 1.5 mol of dinitrophenyl groups per mole subunit can be calculated on the basis of this spectrum—a value that is in good agreement with other experimental data [E. S. J. Arnér, *Biofactors* **10**, 219 (1999); J. Nordberg, L. Zhong, A. Holmgren, and E. S. Arnér, *J. Biol. Chem.* **273**, 10835 (1998)]. The dotted curve in the upper inset shows the differential absorbance of the NADPH-reduced enzyme species. The prominent negative peak in the dashed curve is caused by NADPH consumption in the sample, which is due to the strong oxidase activity of the arylated enzyme.

Tris-HCl, pH 7.4. The large crystals harvested, however, do not yet diffract to a satisfactory resolution. For noncrystallographic applications, for example, biolistic *in vivo* studies of thioredoxin reductase, the crystals are satisfactory.³⁷

³³ M. Bilzer, R. L. Krauth-Siegel, R. H. Schirmer, T. P. Akerboom, H. Sies, and G. E. Schulz, *Eur. J. Biochem.* **138**, 373 (1984).

³⁴ L. Zhong, E. S. J. Arnér, and A. Holmgren, *Proc. Natl. Acad. Sci. U.S.A.* **97**, 5854 (2000).

³⁵ L. Zhong, K. Persson, T. Sandalova, G. Schneider, and A. Holmgren, *Acta Crystallogr. D Biol. Crystallogr.* **56**, 1191 (2000).

³⁶ S. Grömer, in "Die Thioredoxinreduktase von Mensch und Maus—Ein Selenoenzym als Zielmolekül von Chemotherapeutika." MD Thesis. Ruprecht-Karls-Universität, Heidelberg, Germany, 1998.

³⁷ M. A. Keese, R. Saffrich, T. Dandekar, K. Becker, and R. H. Schirmer, *FEBS Lett.* **447**, 135 (1999).

[37] Classification of Plant Thioredoxins by Sequence Similarity and Intron Position

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Thioredoxins are small proteins (~12 kDa) with a cysteine pair present in the almost conserved active site WCGPC, conferring a strong disulfide reductase activity. Our knowledge of plant thioredoxins has been reviewed.¹ One characteristic of plants is the presence of multiple thioredoxin types and of multiple genes for each thioredoxin type in a single genome. All genes are nuclear encoded. This article allows the reader to classify a newly sequenced plant cDNA (or genomic DNA fragment) within a particular type of thioredoxin. The method is based on the analysis of the thioredoxin gene family of the *Arabidopsis thaliana* genome, which is now completely sequenced. The validity of these characteristics to thioredoxin genes of other plants was tested with dicot and monocot sequences available in January 2001.

Arabidopsis Genes Encoding Thioredoxins or Proteins with Thioredoxin Domains

Classic Thioredoxins m, f, and h

The classic plant thioredoxins *m* and *f* were first isolated on a biochemical basis as *in vitro* activators of the chloroplastic enzymes malate dehydrogenase and fructose-1,6-bisphosphatase, respectively. The pea and spinach proteins

¹ P. Schurmann and J. P. Jacquot, *Annu. Rev. Plant Physiol. Plant Mol. Biol.* **51**, 371 (2000).